Real-Time Object Recognition for a Low-Cost Smart Assistance Prototype

Student Research Proposal • Sadril Hossain Ether • Computer Vision • Deep Learning • Robotics

Proposed by	Sadril Hossain Ether
Project type	Student-led research / prototype study
Estimated duration	10–12 weeks

Project Overview

This project explores a lightweight computer-vision prototype that can recognize a small set of everyday objects from a camera feed. The goal is to understand the complete pipeline—from collecting or selecting data to training a model and running inference—while keeping the system simple enough for a student research project.

Student perspective
The project is intentionally scoped so that a small student team can learn, experiment, and produce a useful research artifact without assuming access to a large laboratory or proprietary dataset.

Research Problem

Modern object-detection models can be powerful but may require significant computing resources and careful optimization. For an academic prototype, it is useful to understand the trade-off between detection accuracy, model size, inference speed, and hardware limitations.

Research Objectives

- Select a small set of object classes that are practical for a controlled experiment.
- Prepare and document an appropriately licensed dataset or collect a small consent-based dataset.
- Train or fine-tune a lightweight object-detection model.
- Measure detection accuracy and inference speed on available hardware.
- Explore basic optimization techniques for running the model on a lower-resource device.

Research Questions

RQ1. How does model size affect real-time inference?

RQ2. Which augmentation strategies help with a small dataset?

RQ3. How much accuracy is lost when moving to lightweight models?

RQ4. Can the prototype run at a useful speed on low-cost hardware?

Proposed Methodology

Step 1. Define object classes and prepare training, validation, and test splits.

Step 2. Apply appropriate image preprocessing and augmentation.

Step 3. Train or fine-tune a lightweight detection model using a reproducible configuration.

Step 4. Evaluate using suitable detection metrics and qualitative examples.

Step 5. Measure latency and resource use, then discuss the trade-offs between model accuracy and deployment cost.

Data / Materials / Experimental Resources

The exact resources will be finalized with the collaborator after a feasibility discussion. The preferred approach is to start with public or self-generated material, keep the experiment small enough to reproduce locally, and record the source, preprocessing decisions, assumptions, and version of each important component.

Initial resource plan: Licensed image dataset or consent-based images, training configuration, camera/edge device if available.

Evaluation Plan

Evaluation will be based on measurable criteria rather than only a successful demo. Depending on the project, this may include classification metrics, detection quality, latency, error analysis, reproducibility, or qualitative review. Baselines will be established before more complex techniques are introduced so that the contribution of each change can be discussed clearly.

Reproducibility and Documentation

The team will keep a simple experiment log containing dataset versions, preprocessing choices, model or system settings, evaluation results, and important observations. Code will be organized so that the main experiment can be rerun without relying on undocumented manual steps. Charts and tables will be generated from saved results where possible.

Expected Challenges

- Limited time and computing resources may require a smaller experimental scope.
- Dataset quality may affect conclusions more than model complexity.
- A strong-looking metric may hide weaknesses that appear during error analysis.
- The research question may need refinement after an initial feasibility experiment.

Planned Mitigation

The project will begin with a small baseline and a feasibility check. If data or hardware limitations make the original scope unrealistic, the team will narrow the research question rather than artificially inflating results. Each limitation will be reported as part of the final research discussion.

Expected Outcomes

- A working object-recognition prototype.
- Documented dataset and training pipeline.
- Accuracy and inference-speed evaluation.
- A short study of lightweight deployment trade-offs.

Potential Research Collaboration

I am looking for someone interested in computer vision, deep learning, or robotics who can help with dataset preparation, model selection, or evaluation. A collaborator familiar with edge devices would be especially helpful for the deployment stage.

Technical Stack

Python, PyTorch or TensorFlow, OpenCV, Jupyter Notebook; optional Raspberry Pi or another edge device.

Scope, Ethics, and Responsible Research

Images will be sourced responsibly or collected with appropriate consent. The project will avoid unnecessary personal identification. The prototype is intended for controlled research and will not make safety-critical decisions.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Dataset/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, implementation, experiment design, data analysis, or writing. I would prefer to divide work according to each person's strengths and keep decisions, code, data, and results documented.

Possible Next Step

If a researcher is interested, I would like to first discuss the research question, available resources, and a realistic scope. After that discussion, the proposal can be refined into a clearer study plan with specific responsibilities and milestones.

Note: This document describes a proposed student project. It does not claim completed research, published results, or guaranteed performance. The scope is open to refinement through collaboration.